

70 + 10

1.1.5

$\forall x \in \mathbb{Z}/n\mathbb{Z}, [0] * [x] = [0]$. Assume $\mathbb{Z}/n\mathbb{Z}$ is a group. If $[0]$ is the identity, then $\exists x \in \mathbb{Z}/n\mathbb{Z}, [x] \neq [0], [0] * [x] = [0] \neq [x]$; if $[0]$ is not the identity and $e \neq [0]$ is the identity, $[0] * [0]^{-1} = [0] \neq e$. A contradiction.

1.1.7

$\forall x \in \mathbb{R}, x - 1 < [x] \leq x$, so $\forall x, y \in G, x \star y = x + y - [x + y] \in G$. So $\star$ is a well-defined binary operation on G .

(1) $\forall x, y, z \in G$,

$$\begin{aligned} & (x \star y) \star z \\ &= x + y - [x + y] + z - [x + y - [x + y] + z] \\ &= x + y + z - [x + y] + [x + y] - [x + y + z] \\ &= x + y + z - [y + z] + [y + z] - [x + y + z] \\ &= x \star (y \star z) \end{aligned}$$

(2) 0 is the identity. $\forall x \in G, 0 \star x = x \star 0 = x - [x] = 0$

(3) $\forall x \in G$, the inverse of x is $x^{-1} = 1 - x - [1 - x]$, since

$$\begin{aligned} & x \star x^{-1} \\ &= x + 1 - x - [1 - x] - [x + 1 - x - [1 - x]] \\ &= x + 1 - x - [1 - x] + [1 - x] - [x + 1 - x] \\ &= 1 - [1] \\ &= 0 \end{aligned}$$

(4) $\forall x, y \in G, x \star y = x + y - [x + y] = y + x - [y + x] = y \star x$

So $(G, \star)$ is an Abelian group.

1.1.20

When $|x| < \infty$, $(x^{-1})^{|x|} = (x^{|x|})^{-1} = 1^{-1} = 1$, so $|x^{-1}| \leq |x|$. On the other hand, $|x| = |(x^{-1})^{-1}| \leq |x^{-1}|$. So $|x| = |x^{-1}|$.

When $|x| = \infty$, assume $(x^{-1})^k = 1$ for some $k > 0$, then $x^k = ((x^{-1})^k)^{-1} = 1$, contradict with $|x| = \infty$.

1.1.29

If both A, B are Abelian group, $\forall (a, b), (c, d) \in A \times B$,
 $(a, b) * (c, d) = (a * c, b * d) = (c * a, d * b) = (c, d) * (a, b)$.

If both $A \times B$ is an Abelian group, $\forall a, c \in A$,
 $(a * c, 1) = (a, 1) * (c, 1) = (c, 1) * (a, 1) = (c * a, 1)$, so A is an Abelian group. Same argument for B .

1.1.31

Assume for contradiction that G does not have an element with order 2, then $\forall x \neq 1, x \in G$, we have $x^2 \neq 1$, so $x^{-1} \neq x$. So all the elements in G except 1 form a matching, so $|G|$ is an odd number, a contradiction.

1.2.3

$\forall x \in G$, if x is not a power of r , then $x = sr^i$ for some $0 \leq i < n$. Then $x^{-1} = r^{-i}s^{-1} = r^{-i}s = sr^i = x$, so $x^2 = 1$. Since $x \neq 1$, $|x| = 2$.

1.2.16

$$\begin{aligned}(x_1 y_1)^2 &= 1 \\ \Leftrightarrow x_1 y_1 x_1 y_1 &= 1 \\ \Leftrightarrow x_1 y_1 x_1 &= y_1^{-1} \\ \Leftrightarrow x_1 y_1 x_1 &= y_1 \\ \Leftrightarrow x_1 y_1 &= y_1 x_1^{-1}\end{aligned}$$

So the group is D_4 .

1.1.33

(a) Assume for contradiction that $x^i = x^{-i}$ for some $1 \leq i < n$, then $x^{2i} = 1$. If $2i < n$, $|x| \leq 2i < n$ since $x^{2i} = 1$; if $2i > n$, $|x| \leq 2(n-i) < n$ since $x^{2(n-i)} = x^{2n} x^{-2i} = 1^2 \times 1^{-1} = 1$. This contradicts with $|x| = n$.

(b) For $i = k$, $x^{2i} = 1$, so $x^i = x^{-i}$.

For $1 \leq i < k$, assume $x^i = x^{-i}$, then $x^{2i} = 1$, $|x| \leq 2i < n$, a contradiction.

For $k < i < n$, assume $x^i = x^{-i}$, then $x^{2(n-i)} = x^{2n} x^{-2i} = 1$, $|x| \leq 2(n-i) < n$, a contradiction.